Musical examples

The musical examples, including scores and audios, links to the official Analyzing Classical Form website.

Among the https://acf.research.mcgill.ca/supp_list.phpcomplete list of example we look at how the form inference model perform on only a few of them.

Model output

Identifying presentation

example 2-11

🖼️🔊 **Click to Play**

We distinguish between bindings for idea and bindings for formal unit binding. For the presentation, it is the idea that is repeats, while for the period the repeating thing is the formal unit (ABAC)

```
Input surface:
size = 11
  {a ⟦I None⟧}
  {a ⟦IV None⟧}
```

```
Output analysis:
  size=9 compressionRate=1.2222222222222223
  Seq
    idea = "a"
    parameters = ConcatParam
      ⟦I None⟧
      ⟦IV None⟧
```

```
  size=11 compressionRate=1.0
  Concatenate
    {"a" ⟦I None⟧}
    {"a" ⟦IV None⟧}
```

This example illustrates the basics of how variable binding leads to smaller size. By default, we assign a high cost of raw repetition markers from the input (as they are placeholders for the surface motivic ideas). On the other hand, calling a binding variable just cost 1.

example 2-12

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This time we make the input a little bit harder in that the basic idea is not an atomic marker, but the concatenation of two. And the model need to identify the repetition of “a”+“b” as the basic idea, and correctly build up the phrase parameter for the concatenated phrases.

```
Input surface:
size = 27
  {a ⟦I-V None⟧}
  {b ⟦I-III None⟧}
  {a ⟦VI-II None⟧}
  {b ⟦VI-III None⟧}



Output analysis:
  size=23 compressionRate=1.173913043478261
  Seq
    idea = "a"+"b"
    parameters = ConcatParam
      ⟦I-V None⟧
      ConcatParam
        ⟦I-III None⟧
        ConcatParam
          ⟦VI-II None⟧
          ⟦VI-III None⟧



  size=27 compressionRate=1.0
  Concatenate
    {"a" ⟦I-V None⟧}
    Concatenate
      {"b" ⟦I-III None⟧}
      Concatenate
        {"a" ⟦VI-II None⟧}
        {"b" ⟦VI-III None⟧}
```

Identifying compound themes

example 6-1

🖼️🔊 **Click to Play**

```
Input surface:
size = 60
  {a ⟦I-V None⟧}
  {b ⟦I-V-I-V Weak⟧}
  {c ⟦IV-V-I None⟧}
  {d ⟦VI-VII-I-V Weak⟧}
  {a ⟦I-V None⟧}
  {b ⟦I-V-I-V Weak⟧}
  {c ⟦V None⟧}
  {e ⟦I Strong⟧}




Output analysis:
  size=59 compressionRate=1.0169491525423728
  Period
    antecedentHead = "a"+"b"
    antecedent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ⟦I-V None⟧ ⟦I-V-I-V Weak⟧
                   Concatenate
                     {"c" ⟦IV-V-I None⟧}
                     {"d" ⟦VI-VII-I-V Weak⟧}
    consequent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ⟦I-V None⟧ ⟦I-V-I-V Weak⟧
                   Concatenate
                     {"c" ⟦V None⟧}
                     {"e" ⟦I Strong⟧}



  size=61 compressionRate=0.9836065573770492
  Concatenate
    {"a" ⟦I-V None⟧}
    Concatenate
      {"b" ⟦I-V-I-V Weak⟧}
      Period
        antecedentHead = "c"
        antecedent = Concatenate
                       UseIdeaWith   $0 ⟦IV-V-I None⟧
                       Concatenate
                         {"d" ⟦VI-VII-I-V Weak⟧}
                         Concatenate
                           {"a" ⟦I-V None⟧}
                           {"b" ⟦I-V-I-V Weak⟧}
        consequent = Concatenate
                       UseIdeaWith   $0 ⟦V None⟧
                       {"e" ⟦I Strong⟧}
```

example 6-2

🖼️🔊 **Click to Play**

```
Input surface:
size = 64
  {a ⟦I None⟧}
  {b ⟦I None⟧}
  {c ⟦I None⟧}
  {a ⟦V None⟧}
  {b ⟦V None⟧}
  {c ⟦V None⟧}
  {b ⟦I-VI None⟧}
  {b ⟦IV-II None⟧}
  {b ⟦V None⟧}
  {d ⟦I-II-V-I Strong⟧}



Output analysis:
  size=59 compressionRate=1.0847457627118644
  Concatenate
    Seq
      idea = "a"+"b"+"c"
      parameters = ConcatParam
        ConcatParam
          ⟦I None⟧
          ConcatParam
            ⟦I None⟧
            ⟦I None⟧
        ConcatParam
          ⟦V None⟧
          ConcatParam
            ⟦V None⟧
            ⟦V None⟧
    Sentence
      basicIdea = "b"
      presentation = Concatenate
                       UseIdeaWith   $0 ⟦I-VI None⟧
                       UseIdeaWith   $0 ⟦IV-II None⟧
      continuation = Concatenate
                       {$0 ⟦V None⟧}
                       {"d" ⟦I-II-V-I Strong⟧}



  size=62 compressionRate=1.032258064516129
  Sentence
    basicIdea = "a"+"b"+"c"
    presentation = Concatenate
                     UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ConcatParam   ⟦I None⟧
⟦I None⟧
                     UseIdeaWith   $0 ConcatParam   ⟦V None⟧ ConcatParam   ⟦V None⟧
⟦V None⟧
    continuation = Sentence
                     basicIdea = "b"
```

```
presentation = Concatenate
                 UseIdeaWith   $0 ⟦I-VI None⟧
                 UseIdeaWith   $0 ⟦IV-II None⟧
continuation = Concatenate
                 {$0 ⟦V None⟧}
                 {"d" ⟦I-II-V-I Strong⟧}
```

example 6-5

🖼️🔊 Click to Play

```
Input surface:
size = 92
  {a ⟦I-V None⟧}
  {a ⟦V-I None⟧}
  {b ⟦IV-I None⟧}
  {b ⟦IV-I None⟧}
  {c ⟦II None⟧}
  {c ⟦I None⟧}
  {d ⟦V Weak⟧}
  {a ⟦I-V None⟧}
  {a ⟦V-I None⟧}
  {b ⟦IV-I None⟧}
  {b ⟦IV-I None⟧}
  {c ⟦II None⟧}
  {c ⟦I None⟧}
  {e ⟦V-I Strong⟧}




Output analysis:
  size=80 compressionRate=1.15
  Period
    antecedentHead = Twice "a"+Twice "b"+Twice "c"
    antecedent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ConcatParam   ⟦I-V None⟧
ConcatParam   ⟦V-I None⟧ ⟦IV-I None⟧ ConcatParam   ⟦IV-I None⟧ ConcatParam   ⟦II
None⟧ ⟦I None⟧
                   {"d" ⟦V Weak⟧}
    consequent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ConcatParam   ⟦I-V None⟧
ConcatParam   ⟦V-I None⟧ ⟦IV-I None⟧ ConcatParam   ⟦IV-I None⟧ ConcatParam   ⟦II
None⟧ ⟦I None⟧
                   {"e" ⟦V-I Strong⟧}



  size=81 compressionRate=1.1358024691358024
  Period
    antecedentHead = Twice "a"+Twice "b"
    antecedent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ConcatParam   ⟦I-V None⟧ ⟦V-I None⟧
ConcatParam   ⟦IV-I None⟧ ⟦IV-I None⟧
                   Concatenate
                     Seq
                       idea = "c"
                       parameters = ConcatParam
                         ⟦II None⟧
                         ⟦I None⟧
                     {"d" ⟦V Weak⟧}
    consequent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ConcatParam   ⟦I-V None⟧ ⟦V-I None⟧
ConcatParam   ⟦IV-I None⟧ ⟦IV-I None⟧
```

```
Concatenate
  Seq
    idea = "c"
    parameters = ConcatParam
      ⟦II None⟧
      ⟦I None⟧
  {"e" ⟦V-I Strong⟧}
```

Others

supp 6-1

🖼️🔊 **Click to Play**

example 5-11

🖼️🔊 **Click to Play**

examples from annotation_pilot

```
Input surface:
size = 335
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {C ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {E ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {C ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {E ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {H ⟦I None⟧}
  {I ⟦I None⟧}
  {J ⟦I None⟧}
  {B ⟦I Weak⟧}
  {K ⟦I None⟧}
  {K ⟦I None⟧}
  {L ⟦I None⟧}
  {M ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {N ⟦I None⟧}
  {G' ⟦I Strong⟧}
  {H' ⟦I None⟧}
  {H' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {H ⟦I None⟧}
  {I ⟦I None⟧}
  {J ⟦I None⟧}
  {B ⟦I Weak⟧}
  {K ⟦I None⟧}
```

```
  {K ⟦I None⟧}
  {L ⟦I None⟧}
  {M ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {N ⟦I None⟧}
  {G' ⟦I Strong⟧}
  {H' ⟦I None⟧}
  {H' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}



Output analysis:
  size=146 compressionRate=2.2945205479452055
  SimpleBinary
    aPart = Period
              antecedentHead = "A"+"B"
              antecedent = Concatenate
                             UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                             Concatenate
                               {"C" ⟦I None⟧}
                               {"D" ⟦I Weak⟧}
              consequent = Concatenate
                             UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                             HybridAntCont
                               antecedent = Concatenate
                                              {"E" ⟦I None⟧}
                                              {"D" ⟦I Weak⟧}
                               continuation = Concatenate
                                                TwiceFormalUnit
                                                  {"A'" ⟦I None⟧}
                                                Concatenate
                                                  {"F" ⟦I None⟧}
                                                  {"G" ⟦I Strong⟧}
    bPart = Concatenate
              {"H" ⟦I None⟧}
              Concatenate
                {"I" ⟦I None⟧}
                Concatenate
                  {"J" ⟦I None⟧}
                  Concatenate
                    Concatenate
                      {"B" ⟦I Weak⟧}
                      Concatenate
                        TwiceFormalUnit
                          {"K" ⟦I None⟧}
                        Concatenate
                          {"L" ⟦I None⟧}
                          {"M" ⟦I Weak⟧}
                    Concatenate
                      TwiceFormalUnit
                        {"A'" ⟦I None⟧}
                      Concatenate
```

```
                      Concatenate
                        {"N" ⟦I None⟧}
                        {"G'" ⟦I Strong⟧}
                      Concatenate
                        TwiceFormalUnit
                          {"H'" ⟦I None⟧}
                        Concatenate
                          {"F" ⟦I None⟧}
                          {"G" ⟦I Strong⟧}



size=148 compressionRate=2.2635135135135136
Concatenate
  TwiceFormalUnit
    Period
      antecedentHead = "A"+"B"
      antecedent = Concatenate
                     UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                     Concatenate
                       {"C" ⟦I None⟧}
                       {"D" ⟦I Weak⟧}
      consequent = Concatenate
                     UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                     HybridAntCont
                       antecedent = Concatenate
                                      {"E" ⟦I None⟧}
                                      {"D" ⟦I Weak⟧}
                       continuation = Concatenate
                                        TwiceFormalUnit
                                          {"A'" ⟦I None⟧}
                                        Concatenate
                                          {"F" ⟦I None⟧}
                                          {"G" ⟦I Strong⟧}
  TwiceFormalUnit
    Concatenate
      {"H" ⟦I None⟧}
      Concatenate
        {"I" ⟦I None⟧}
        Concatenate
          {"J" ⟦I None⟧}
          Concatenate
            Concatenate
              {"B" ⟦I Weak⟧}
              Concatenate
                TwiceFormalUnit
                  {"K" ⟦I None⟧}
                Concatenate
                  {"L" ⟦I None⟧}
                  {"M" ⟦I Weak⟧}
            Concatenate
              TwiceFormalUnit
                {"A'" ⟦I None⟧}
              Concatenate
                Concatenate
                  {"N" ⟦I None⟧}
```

```
  {"G'" ⟦I Strong⟧}
Concatenate
  TwiceFormalUnit
    {"H'" ⟦I None⟧}
  Concatenate
    {"F" ⟦I None⟧}
    {"G" ⟦I Strong⟧}
```

```
Input surface:
size = 623
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {k ⟦IV None⟧}
  {g ⟦II None⟧}
  {k ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
```

```
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {k ⟦IV None⟧}
  {g ⟦II None⟧}
  {k ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}



Output analysis:
  size=148 compressionRate=4.20945945945946
```

```
  Ternary
    exposition = SimpleBinary
                  aPart = HybridAntCont
                            antecedent = Sentence
                                           basicIdea = "a"
                                           presentation = Concatenate
                                                            UseIdeaWith   $0 ⟦I
None⟧
                                                            UseIdeaWith   $0 ⟦V
None⟧
                                           continuation = Concatenate
                                                            {$0 ⟦I None⟧}
                                                            {"b" ⟦V Weak⟧}
                            continuation = Concatenate
                                             TwiceFormalUnit
                                               {"c" ⟦I None⟧}
                                             Concatenate
                                               {"d" ⟦I None⟧}
                                               {"e" ⟦I Strong⟧}
                  bPart = HybridAntCont
                            antecedent = Concatenate
                                           Seq
                                             idea = "a"
                                             parameters = ConcatParam
                                               ⟦I None⟧
                                               ⟦V None⟧
                                           Concatenate
                                             {"f" ⟦V None⟧}
                                             {"b" ⟦V Weak⟧}
                            continuation = Concatenate
                                             TwiceFormalUnit
                                               {"c" ⟦I None⟧}
                                             Concatenate
                                               {"d" ⟦V None⟧}
                                               {"e" ⟦I Strong⟧}
    contrastingMiddile = RoundedBinary
                           exposition = Concatenate
                                          Seq
                                            idea = "g"+"h"
                                            parameters = ConcatParam
                                              ConcatParam
                                                ⟦I None⟧
                                                ⟦I None⟧
                                              ConcatParam
                                                ⟦V None⟧
                                                ⟦V None⟧
                                          Concatenate
                                            Concatenate
                                              {"g" ⟦I None⟧}
                                              {"h" ⟦IV None⟧}
                                            Concatenate
                                              {"i" ⟦V None⟧}
                                              {"j" ⟦I Strong⟧}
                           contrastingMiddile = Seq
                                                    idea = "g"+"k"
                                                    parameters = ConcatParam
```

```
                                                    ConcatParam
                                                      ⟦I None⟧
                                                      ⟦IV None⟧
                                                    ConcatParam
                                                      ⟦II None⟧
                                                      ⟦V None⟧



size=195 compressionRate=3.194871794871795
Concatenate
  TwiceFormalUnit
    HybridAntCont
      antecedent = Sentence
                     basicIdea = "a"
                     presentation = Concatenate
                                      UseIdeaWith   $0 ⟦I None⟧
                                      UseIdeaWith   $0 ⟦V None⟧
                     continuation = Concatenate
                                      {$0 ⟦I None⟧}
                                      {"b" ⟦V Weak⟧}
      continuation = Concatenate
                       TwiceFormalUnit
                         {"c" ⟦I None⟧}
                       Concatenate
                         {"d" ⟦I None⟧}
                         {"e" ⟦I Strong⟧}
  Ternary
    exposition = TwiceFormalUnit
                   HybridAntCont
                     antecedent = Concatenate
                                    Seq
                                      idea = "a"
                                      parameters = ConcatParam
                                        ⟦I None⟧
                                        ⟦V None⟧
                                    Concatenate
                                      {"f" ⟦V None⟧}
                                      {"b" ⟦V Weak⟧}
                     continuation = Concatenate
                                      TwiceFormalUnit
                                        {"c" ⟦I None⟧}
                                      Concatenate
                                        {"d" ⟦V None⟧}
                                        {"e" ⟦I Strong⟧}
    contrastingMiddile = Concatenate
                           RoundedBinary
                             exposition = Concatenate
                                            Seq
                                              idea = "g"+"h"
                                              parameters = ConcatParam
                                                ConcatParam
                                                  ⟦I None⟧
                                                  ⟦I None⟧
                                                ConcatParam
                                                  ⟦V None⟧
```

```
                                        ⟦V None⟧
                                    Concatenate
                                      Concatenate
                                        {"g" ⟦I None⟧}
                                        {"h" ⟦IV None⟧}
                                      Concatenate
                                        {"i" ⟦V None⟧}
                                        {"j" ⟦I Strong⟧}
                          contrastingMiddile = Seq
                                             idea = "g"+"k"
                                             parameters = ConcatParam
                                               ConcatParam
                                                 ⟦I None⟧
                                                 ⟦IV None⟧
                                               ConcatParam
                                                 ⟦II None⟧
                                                 ⟦V None⟧
                        TwiceFormalUnit
                          HybridAntCont
                            antecedent = Sentence
                                           basicIdea = "a"
                                           presentation = Concatenate
                                                            UseIdeaWith   $0 ⟦I
None⟧
                                                            UseIdeaWith   $0 ⟦V
None⟧
                                           continuation = Concatenate
                                                            {$0 ⟦I None⟧}
                                                            {"b" ⟦V Weak⟧}
                            continuation = Concatenate
                                             TwiceFormalUnit
                                               {"c" ⟦I None⟧}
                                             Concatenate
                                               {"d" ⟦I None⟧}
                                               {"e" ⟦I Strong⟧}
```